SMART INDIA HACKATHON 2026 — PRODUCTION-QUALITY PROTOTYPE

CRIMENET AI

AI-Powered Criminal Network Analysis & Investigation Intelligence Platform

Document: Technical Specification & Evaluation Guide **Classification:** Law Enforcement Decision Support

Team: PCE SIH 2026 **Version:** 1.0 (Production)

1 Project Overview

Crimenet is a decision-support and investigation-analysis system engineered for law enforcement agencies and intelligence analysts. It transforms fragmented, multi-source crime data (telecom CDRs, FIR filings, transport registry, financial ledgers, and surveillance sightings) into an interactive knowledge graph.

> ⚠ **MANDATORY ANALYTICAL SAFEGUARD — CRITICAL LEGAL NOTICE**
> **CRIMENET AI is strictly an investigative decision-support platform.** The system never declares guilt or criminality automatically. All findings are classified as analytical leads, anomalies, or structural relationships requiring human investigator verification.

2 Technology Stack

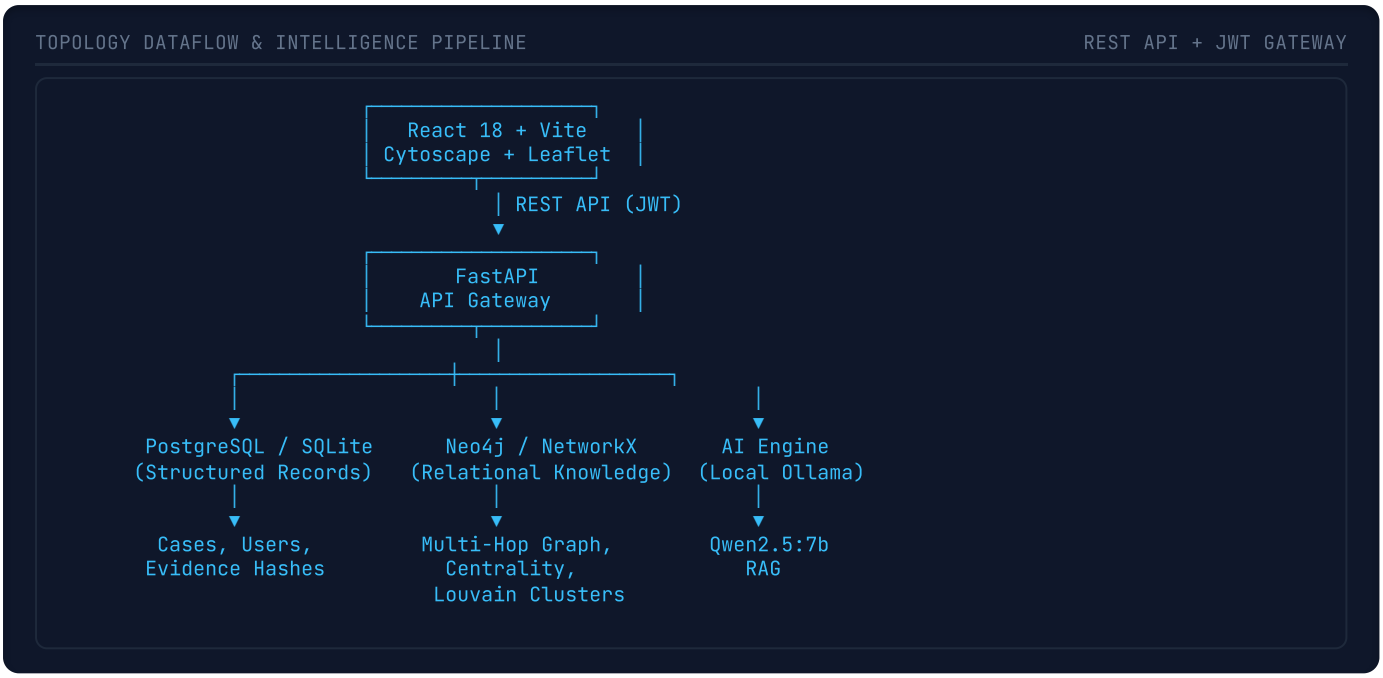
FRONTEND
React.js 18, Vite, Tailwind CSS, Cytoscape.js, Leaflet + OpenStreetMap, Recharts, Lucide React, Axios, React Router.

BACKEND
Python 3.11+, FastAPI, Uvicorn, SQLAlchemy, Pydantic V2, python-jose, bcrypt, ReportLab (PDF compiler).

RELATIONAL & GRAPH DATABASES
PostgreSQL (primary) with automatic local SQLite fallback. **Neo4j** (bolt driver) with an integrated in-memory NetworkX graph fallback.

AI & NLP ENGINE
Local Ollama service (`qwen2.5:7b`), configurable model switcher (Llama3, Mistral, Gemma, Qwen3), regex entity extractor, multi-attribute normalizer, and grounded RAG query builder.

3 High-Level Architecture



4 Key Capabilities & Features

1. Synthetic Investigation Datasets
100 persons, 50 heavy vehicles, 75 phone lines, 30 locations, 20 corporate entities, 100 FIR documents, 500 CDR communications, 300 transactions, 200 events.

2. Flagship Demo Investigation (CASE-2026-001)
Operation Hawkeye with 25 persons, central hub P001 (17 connections), bridge node P014, and verifiable paths.

3. Interactive Knowledge Graph
Cytoscape.js with degree-weighted node sizing, cluster colors, zoom/pan/filter, and PNG export.

4. Graph Centrality Analytics
Degree centrality, Betweenness centrality (bridge detection), and PageRank calculations.

5. Community Detection
Modularity clustering dividing entities into distinct analytical clusters (Cluster A, Cluster B, etc.).

6. Shortest Path Traversal
Discovers & visualizes indirect connection chains between entities (e.g. P001 → P014 → P023) citing supporting records.

7. Explainable Anomaly Detection
- Communication surge: 4.8x baseline volume alert.
- Transaction velocity: INR 850,000 structured outflow.
- Network anomaly: Rapid cross-cluster bridge formation.

8. AI Investigator Assistant (Grounded RAG)
Local LLM answering natural-language inquiries with strict separation of Observations, Analytical Inferences, Uncertainties, and Record Citations.

9. Cryptographic Evidence Integrity
SHA-256 signatures generated upon document upload with live verification badges.

10. Automated PDF Reports & Dossiers
Compiles official intelligence dossiers with topology metrics, anomalies, notes, and statutory disclaimers.

11. Immutable Audit Trail
Tracks all user logins, graph queries, AI prompts, and evidence verifications for complete legal compliance.

5 Demonstration Credentials

The platform provides pre-seeded accounts and 1-click role demo buttons on the login screen:

Role	Username	Password	Demonstration Purpose
INVESTIGATOR	`investigator`	`investigator123`	Case management, graph exploration, AI queries
SENIOR ANALYST	`analyst`	`analyst123`	Anomaly confirmation and hypothesis testing
ADMINISTRATOR	`admin`	`admin123`	System oversight and case registration
AUDITOR / VIEWER	`viewer`	`viewer123`	Read-only compliance and audit verification

6 Official 10-Step SIH Evaluation Workflow

Execute this exact sequence during jury evaluation and presentation:

1. **Open Dashboard (`/dashboard`):** Review metric cards, FIR timeline area chart, entity breakdown, and anomaly queue.
2. **Open Flagship Case (`/cases/CASE-2026-001`):** Click *Launch Case Investigation* to enter Operation Hawkeye.
3. **Interactive Network Graph:** Inspect 54 relationships. Filter by relationship type (e.g., `CALLED` , `OWNS`).
4. **Select Entity P001:** Click Rajesh Sharma to open the side dossier showing 17 connections, betweenness 0.42, 3 vehicles, and 3 phones.
5. **Shortest Path Finder:** Click *Trace Graph Path* from P001 to P023. Observe multi-hop connection via bridge node P014 with supporting records `FIR-1023` and `CDR-1045` .
6. **Temporal Timeline (`/timeline`):** Inspect chronological development of transit sightings, telecom bursts, and financial transfers.
7. **Alert Center (`/alerts`):** Review the 4.8x communication surge alert for P014 and execute *Confirm Lead* analyst workflow.
8. **AI Assistant Reasoning (`/ai-assistant`):** Ask *"Summarize the important relationships in this case and cite supporting records."* View grounded observations and citations without hallucinations.
9. **Evidence Integrity (`/documents`):** Upload a test file, observe instant SHA-256 hash generation, and click *INTEGRITY VERIFIED*.
10. **PDF Dossier Export (`/reports`):** Click *Generate Official Investigation PDF Report* and download the styled court-admissible dossier.

7 Local Setup & Execution Guide

Prerequisites: Python 3.11+, Node.js 18+ and npm. *(Optional: Docker & Docker Compose, Ollama)*

Method A: Native Local Execution (Fastest for Demo)

1. Backend Setup

```
# Navigate to backend directory
cd backend
# Create virtual environment
python -m venv .venv
source .venv/bin/activate
# Install & seed database
pip install -r requirements.txt
python -m app.utils.seed
# Start FastAPI backend
uvicorn app.main:app --reload
```

Swagger Docs: `http://localhost:8000/docs`

2. Frontend Setup

```
# Navigate to frontend
cd frontend
# Install and start Vite dev server
npm install
npm run dev
```

Application UI: `http://localhost:5173`

Method B: Docker Compose

```
# Start all 5 microservices (Postgres, Neo4j, Ollama, Backend, Frontend)
docker compose up -d
docker exec -it crimenet_ollama ollama pull qwen2.5:7b
```

8 Ollama Local LLM Configuration

To connect real local LLM inference:

1. Install Ollama | 2. Run: `ollama pull qwen2.5:7b` | 3. Start: `ollama serve`

Note: If Ollama is offline or uninstalled, CRIMENET AI automatically activates its high-performance local heuristic rule engine, ensuring 100% demo uptime without crashes.

9 Test Suite Verification

```
cd backend
pytest tests/test_api.py -v
```

All 7 critical integration tests verify: JWT authentication & route security, Knowledge graph structure & edge connectivity, Shortest path multi-hop resolution, Centrality metrics (Degree, Betweenness, PageRank), AI safety guardrails (prohibition of guilt declarations), and Cryptographic SHA-256 evidence integrity verification.

10 Statutory Safeguards & Ethical AI

- **Strictly Decision-Support:** AI outputs are framed as analytical leads requiring field verification.
- **No Hallucinations:** RAG prompt structures ensure only retrieved database and graph IDs are cited.
- **Tamper-Evident:** All investigations maintain an immutable SHA-256 cryptographic chain of custody.
- **No Real-World Data:** All names, telephone numbers, and addresses are synthetically generated.